Instituto Superior de Engenharia
Politécnico de Coimbra

Advanced Programming

2025/2026

Practical Work

Final Stage – Sounds, Delivery, and Final Presentation

1. Objectives

In this final stage, the main focus will be on refining and finalizing the entire game logic, as well as incorporating additional user interaction elements. As this is the final submission, this stage includes writing the final report and preparing the presentation/demonstration of the developed game.

2. Requirements

In the context of this final stage, sounds must be incorporated into the application. The sound files must be stored in the folder already provided for this purpose in the repository (`resources/sounds`) and managed through a `SoundManager` class in the `pt.isec.pa.deepsea.ui.res` *package*. The application must generate sounds for every movement of the drone or ship and provide appropriate sounds for special events, such as when the drone overlaps with an animal or when it is destroyed. Note that the system should allow two sounds to play consecutively, and the application must properly manage the sequence of these sounds. For example, when moving the drone, if the user moves very quickly, the sound of a new movement should cancel out the previous movement sound. If the movement involves passing over an animal, the movement sound must play first, and only after it finishes should the sound for overlapping with the animal play. There must be an option to turn the application sounds on or off, managed through a *JavaFX Beans boolean property* within the `SoundManager` class.

In the context of this stage, you must also ensure that the grids resize proportionally to the window size whenever the user resizes the window.

Other tasks to be completed during this stage:

- Finalization of unit tests for the main data model classes;
- Preparation of the application to display a second window, identical to the one containing the user interface introduced in the third stage. The content of both windows must always sync with the game state, and either window can be used by the user to interact with the game.
- Finalization of the *Javadoc* documentation for the main classes;
- Completion of the final report, which must include:
    - A concise explanation of the game, clarifying the choices made by the group;
    - The state diagram;
    - The class diagram;
    - An explanation of the UI design choices;
    - A summary table of planned tasks, showing those that were completed and those that were not;

- Preparation of the final presentation/demonstration – project defenses will take place during the last week of classes, with some being held during regular class hours. Some defenses will need to be scheduled for extra slots during that week or at the beginning of the following week (June 8th and 9th). Groups must prepare a demonstration of all application features, which cannot exceed 10 minutes. For this purpose, you may use the defense mode mentioned in previous stages and/or create a new defense mode tailored to this demonstration.

3. Notes on carrying out the stage

Submission deadline: ***2026.06.01 11:59 am***

Management and submission: *GitHub+Classroom*, using the repository created for each group.

Branch name: ***StageFinal***

By the deadline the code should be integrated into the main *branch* via ***Squash and Merge***. Each group must have only one *Pull Request* for the step's branch and the Squash & Merge must be done on *the GitHub* website.

They must not change or delete any *branch* that has been created and submitted for evaluation, other than in the presence of a teacher of the curricular unit.

Students should define rules for making *commits* in each group that allow to reflect a balanced distribution of tasks among the members of the groups (in other words, we don't want to see one member of the group making 20 *commits* to change prints or readme files and the other making only one *commit* corresponding to the creation of 20 classes). Students who are doing the work individually should also make regular *commits* and not just one per step.